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SYMPOSIUM

Temperature-dependent Developmental Plasticity and Its Effects on Allen's and Bergmann's Rules in Endotherms

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Synopsis Ecogeographical rules, describing common trends in animal form across space and time, have provided key insights into the primary factors driving species diversity on our planet. Among the most well-known ecogeographical rules are Bergmann's rule and Allen's rule, with each correlating ambient temperature to the size and shape of endotherms within a species. In recent years, these two rules have attracted renewed research attention, largely with the goal of understanding how they emerge (e.g., via natural selection or phenotypic plasticity) and, thus, whether they may emerge quickly enough to aid adaptations to a warming world. Yet despite this attention, the precise proximate and ultimate drivers of Bergmann's and Allen's rules remain unresolved. In this conceptual paper, we articulate novel and classic hypotheses for understanding whether and how plastic responses to developmental temperatures might contributed to each rule. Next, we compare over a century of empirical literature surrounding Bergmann's and Allen's rules against our hypotheses to uncover likely avenues by which developmental plasticity might drive temperature-phenotype correlations. Across birds and mammals, studies strongly support developmental plasticity as a driver of Bergmann's and Allen's rules, particularly with regards to Allen's rule. However, plastic contributions toward each rule appear largely non-linear and dependent upon: (1) efficiency of energy use (Bergmann's rule) and (2) thermal advantages (Allen's rule) at given ambient temperatures. These findings suggest that, among endotherms, rapid changes in body shape and size will continue to co-occur with our changing climate, but generalizing the direction of responses across populations is likely naive.

Introduction

Phenotypic variation, both within and among species, is a key contributor to the beauty and resilience of life. In their theories of evolution, both Darwin and Wallace recognized this importance of variation (Wallace 1855; Darwin 1859, 1868) but lacked a formal understanding of how it might first arise. However, Darwin speculated that traits within individuals—or otherwise identical individuals—were likely malleable and varied according to environmental context (reviewed in Winther 2000). Today, this speculated process is best known as "phenotypic plasticity" and is widely understood as a primary strategy to cope with, or even exploit, novel or changing environments (see, for example, Bradshaw 1965; West-Eberhard 1989; Brooker et al. 2022).

Some of the most striking displays of phenotypic plasticity occur in response to temperature. In the Chinese primrose (*Primula sinensis*), flowers that develop red at 20°C emerge white at 30°C, regardless of parentage (Baur 1919). Similarly, five-spotted hawkmoth larvae (*Manduca quinquemaculata*) from the same brood develop black when raised at a mild ambient temperature (<20°C) but bright green when raised in the warmth (>28°C; Suzuki and Nijhout 2006). In fish, Atlantic halibut (*Hippoglossus hippoglossus*) raised in

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